Healthcare Operations & Patient Analytics Dashboard

Hospital admissions, outpatient appointments, no-shows, resource use and billing

1. Business Problem Statement

Hospital management needs one clear view of patient activity, appointment reliability, department workload, treatment outcomes and billing. Today these sit in separate records, so it is hard to see where patients wait too long, where appointments are lost to no-shows, and which areas carry the most workload and cost. Every missed appointment is an unused doctor slot, and long waits make patients less likely to attend.

2. Core Business Objectives

- Measure the volume of patients, admissions and appointments, and how they change over time.
- Find which departments, doctors and patient groups carry the highest workload.
- Find where and why appointments are missed (no-shows) and how waiting time and reminders affect this.
- Compare treatment outcomes and the costs behind them.
- Track resource use (bed-days, length of stay, rooms) and billing revenue.
- Give management clear recommendations to cut waiting time, reduce no-shows and use capacity better.

3. Data Used

Two public Kaggle datasets were used. They have no common key, so they are analysed as two linked views, not joined.

Dataset	Rows	Used for
Healthcare Dataset (prasad22)	55,500	Admissions, doctors, diagnosis, billing, stay, outcome
Medical Appointment No Shows	110,527	Appointments, waiting time, no-shows, patient groups

Important limits: (1) the admissions data has no department column, so each medical condition is mapped to a department (for example Cancer to Oncology). (2) The appointments data has no doctor, department or billing. (3) Neither dataset records cancellations, so the no-show flag is the only missed-appointment measure. (4) The admissions data is synthetic, so most of its groups are almost identical; the report says so where it matters.

4. Data Cleaning, Transformation and Preparation

Dataset	Step	Rows
Admissions	Removed exact duplicate rows	534
Admissions	Fixed random upper/lower case in names	all
Admissions	Removed negative billing amounts	106
Admissions	Converted dates, added length of stay	all
Admissions	Added age group, department, floor, band, outcome	all
Appointments	Renamed misspelled columns	all
Appointments	Removed impossible ages (below 0 or above 100)	8
Appointments	Removed bookings after the appointment date	5
Appointments	Added wait days, wait band, age group, no-show flag	all

Result: 54,860 clean admissions (from 55,500) and 110,514 clean appointments (from 110,527). Neither dataset had missing values. Waiting time is counted in whole days between the booking date and the appointment date.

The prepared tables are saved in Healthcare_Clean_Data.xlsx: Admissions, Appointments, two date tables, a department table, the cleaning summary, the DAX measures and a data dictionary.

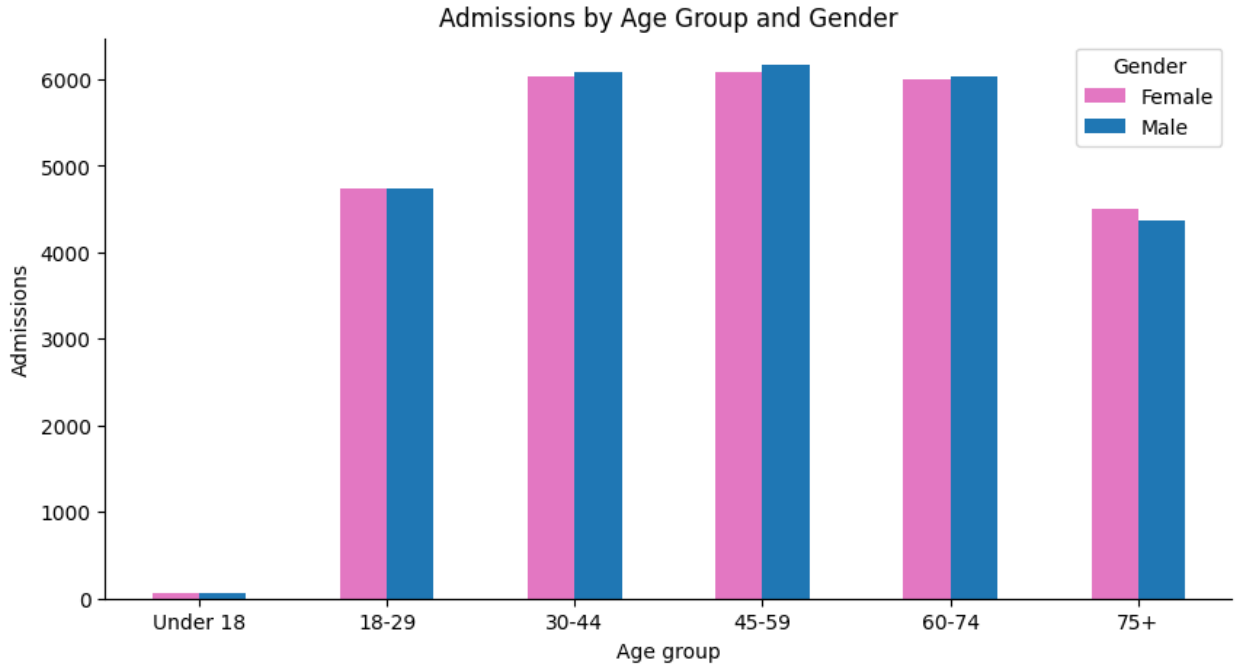
5. KPI Snapshot

Admissions KPI	Value	Appointments KPI	Value
Total admissions	54,860	Total appointments	110,514
Total billing revenue	1,404.1 M	Unique patients	62,294
Average bill	25,595	No-show rate	20.2%
Average stay	15.5 days	Average wait	10.2 days
Normal test result rate	33.4%	Same-day appointments	34.9%

The two datasets cover different periods: admissions run from May 2019 to May 2024 and appointments from 29 April to 8 June 2016. Trend charts therefore describe each dataset on its own.

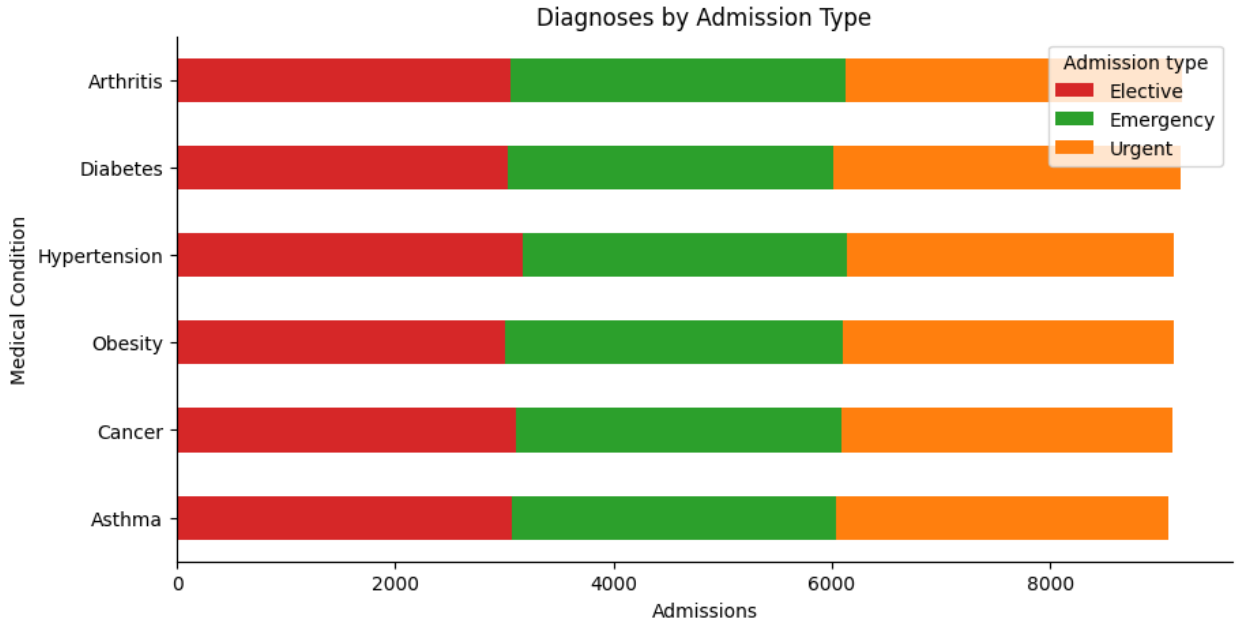
6. Patient Demographics and Segmentation

Which demographics drive hospital utilisation?



Patients aged 30 to 74 make up most admissions, and about 44% of all admissions are aged 45 to 74. Female and male admissions are almost equal in every age group. Under-18 admissions are very rare (116 in total), so this hospital data is an adult population.

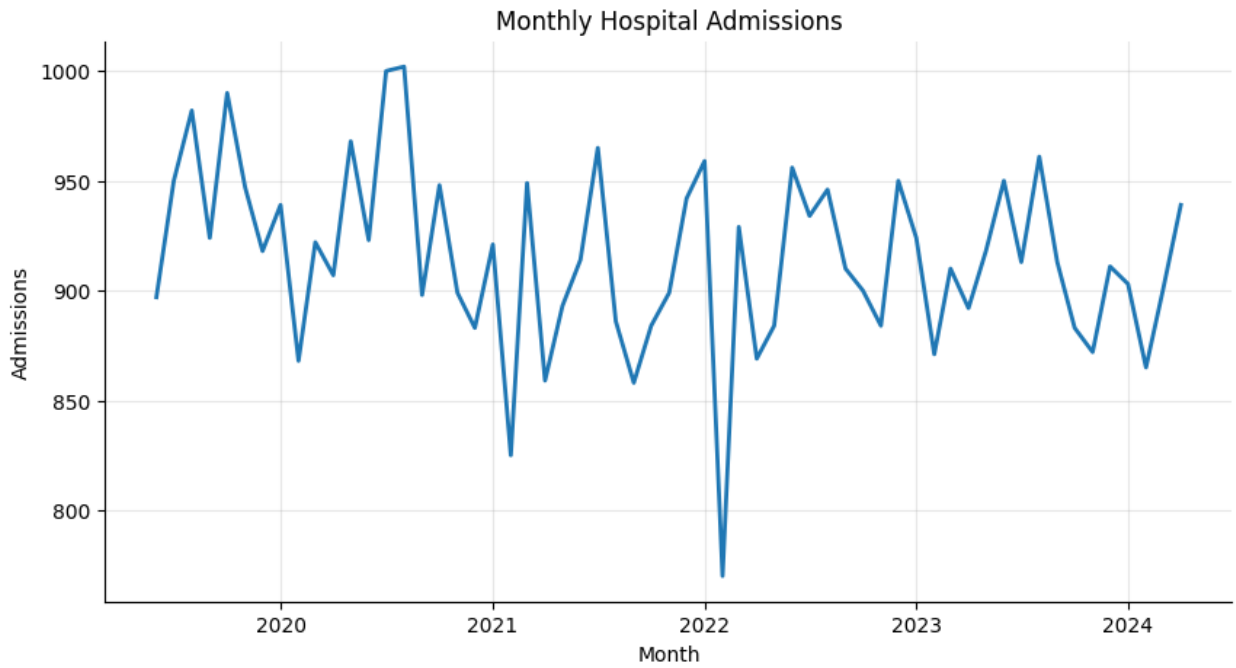
What are the most common diagnoses, and how do patients arrive?



All six diagnoses have a similar volume (about 9,100 to 9,200 each) and each splits roughly one third elective, one third emergency and one third urgent. No single diagnosis stands out as a capacity problem.

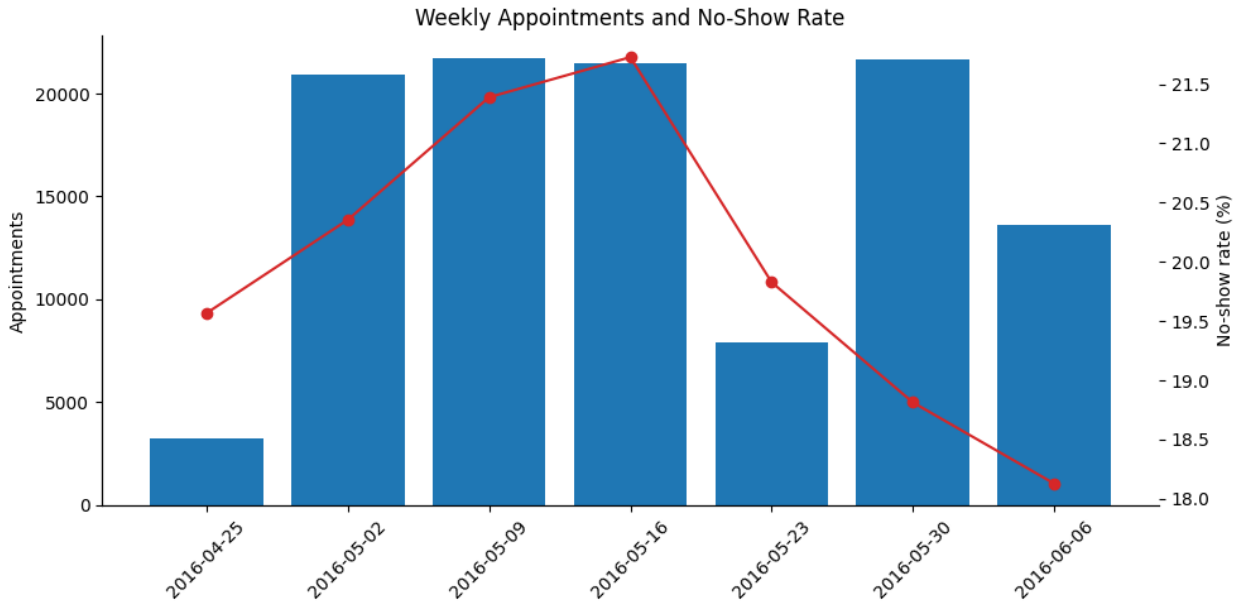
7. Appointment and Hospital Visit Trends

What is the monthly trend in admissions?



Admissions stay in a narrow band around 915 per month. The busiest month was August 2020 (1,002) and the quietest February 2022 (770). There is no strong long-term growth or seasonal pattern.

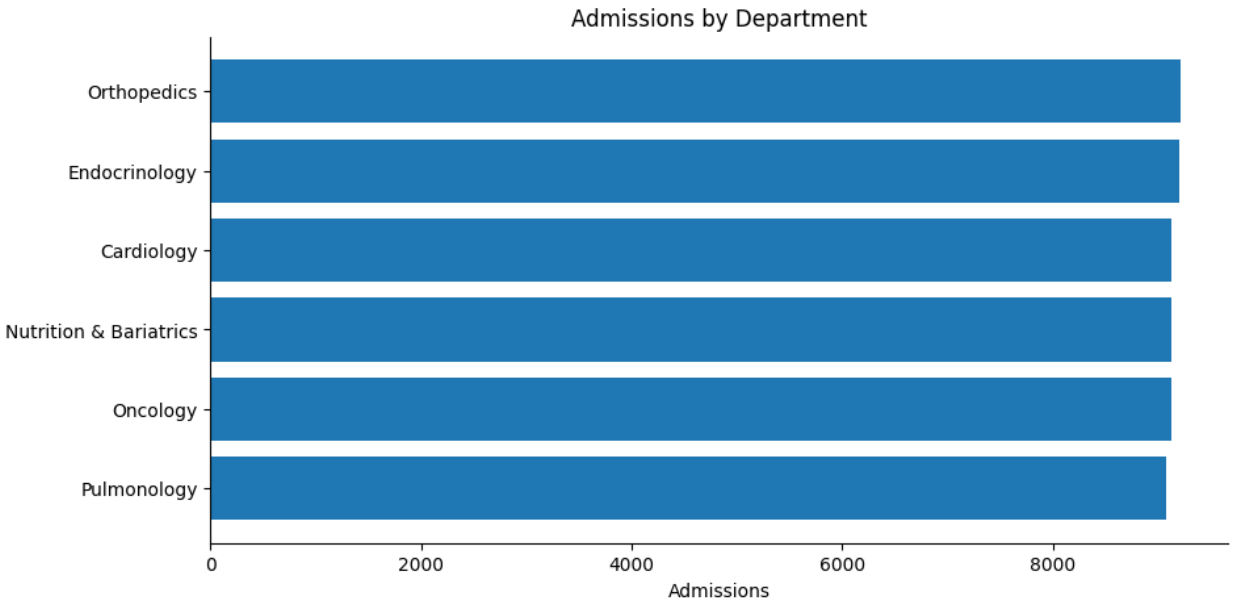
How do weekly appointments and the no-show rate move?



Weekly volume is steady at about 21,000 appointments, apart from partial weeks at the start, end and one holiday week. The no-show rate stays between 18% and 22%, with the lowest rate (18.1%) in the final week.

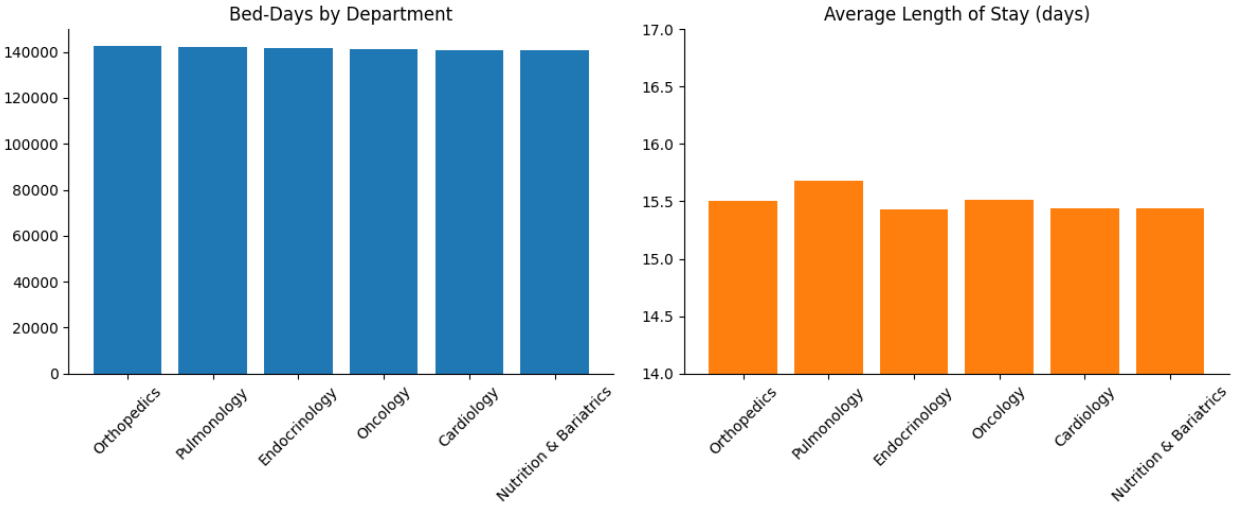
8. Department-wise Performance

Which departments receive the most patients?



Orthopedics (9,207) and Endocrinology (9,197) lead, but the gap to the lowest department (Pulmonology, 9,077) is only 1.4%. Patient volume is evenly spread, so department differences must be judged on stay length and outcomes, not on volume.

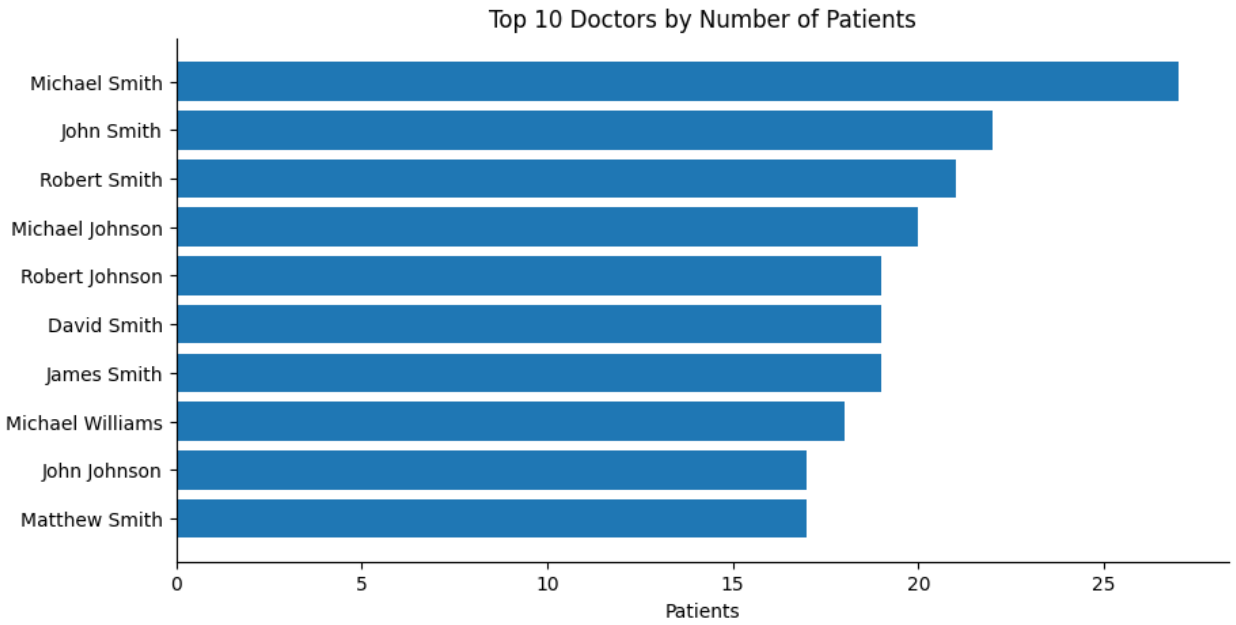
Which departments carry the highest workload?



Orthopedics has the most bed-days (142,743) and Pulmonology has the longest average stay (15.7 days). All departments are within about 1.3% of each other on bed-days.

9. Doctor-wise Patient and Appointment Analysis

Which doctors handle the most patients?

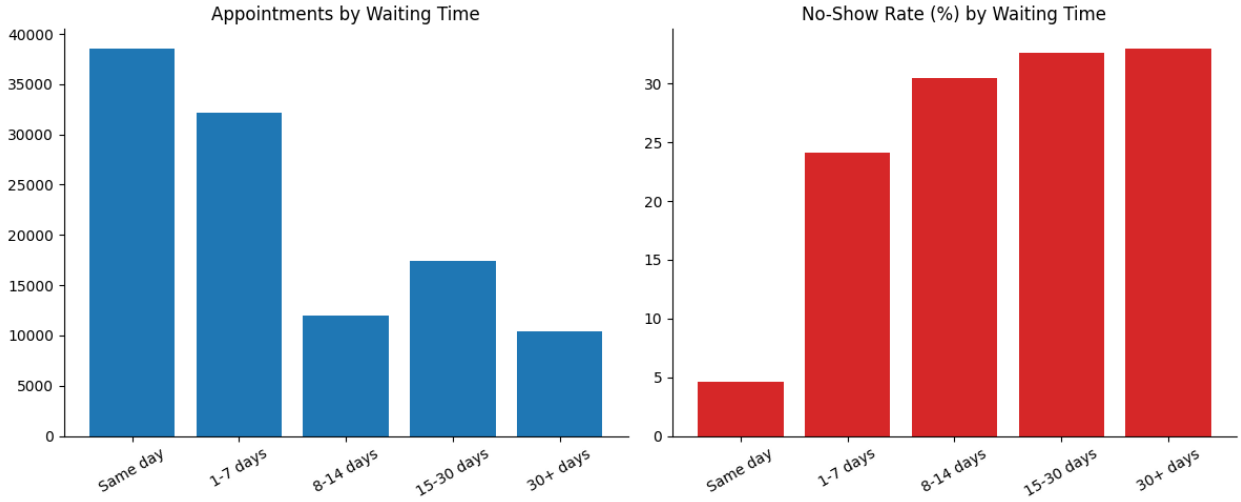


Michael Smith has the most admissions (27, about 784,502 in billing), followed by John Smith (22) and Robert Smith (21). However, the file has 40,276 doctor names and 30,892 of them appear only once, so the doctor field is mostly unique per row. Doctor ranking is shown in the dashboard, but conclusions about individual doctors should be treated with care. The appointments dataset has no doctor field, so appointment analysis by doctor is not possible.

10. Appointment Cancellation and No-Show Analysis

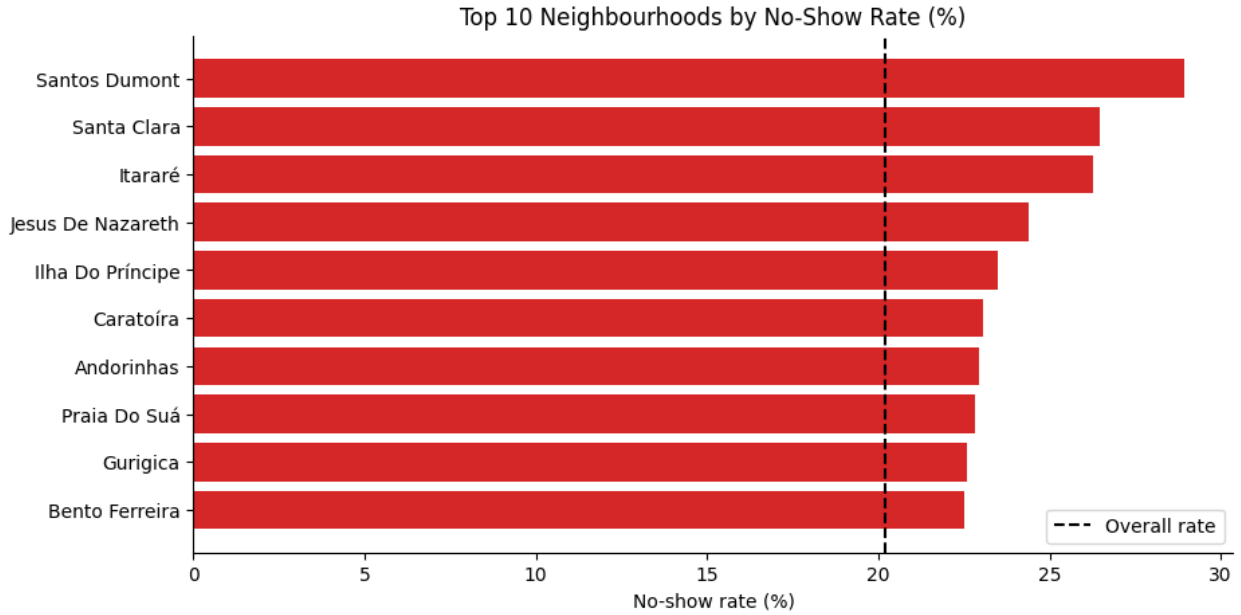
The dataset has no cancellation field, so a no-show (patient booked but did not attend) is the missed-appointment measure. The overall no-show rate is 20.2%.

How does waiting time affect no-shows?



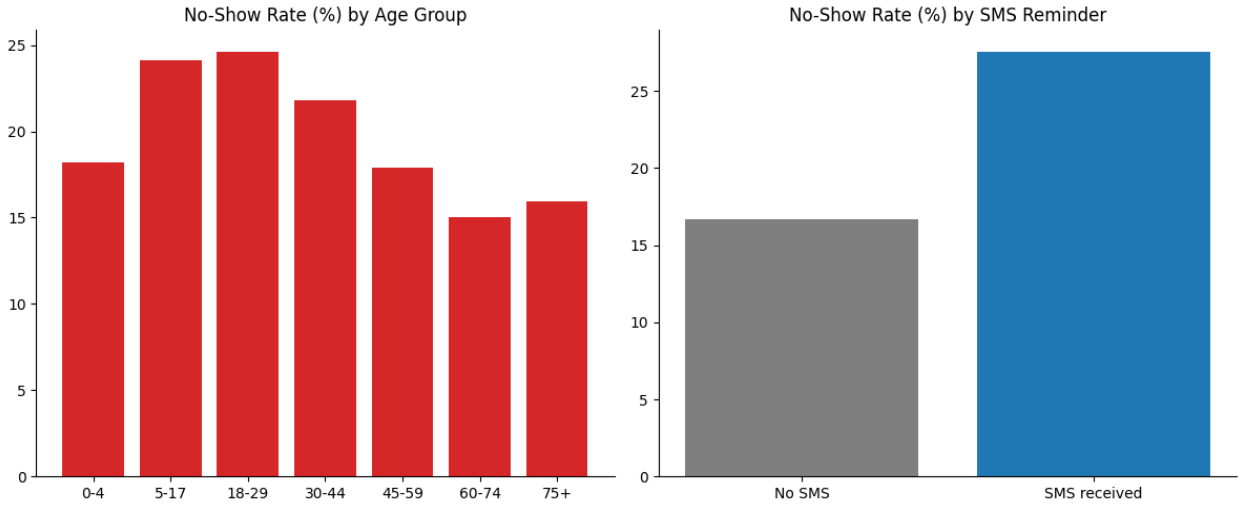
This is the strongest finding in the project. Same-day appointments have a 4.6% no-show rate. It rises to 24.1% for 1 to 7 days, 30.5% for 8 to 14 days, 32.6% for 15 to 30 days and 33.0% beyond 30 days. 65% of appointments are booked at least one day ahead, so waiting time is the main cause of lost slots.

Which neighbourhoods have the highest no-show rates?



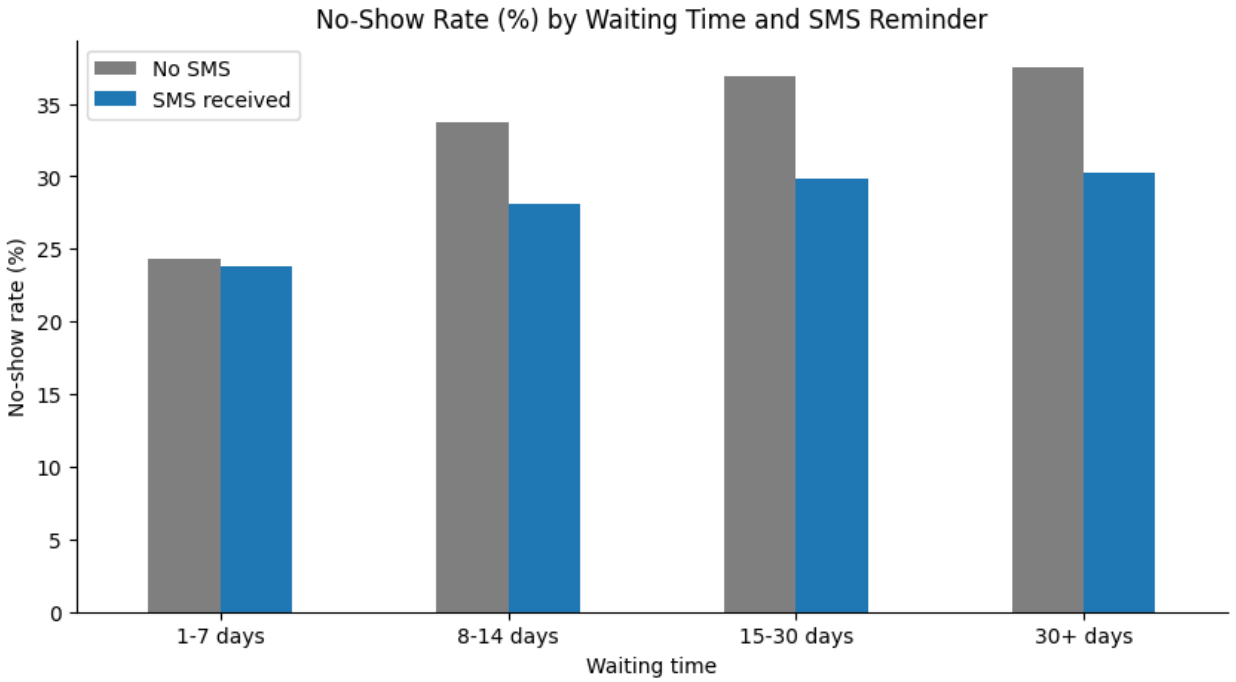
Santos Dumont (28.9%), Santa Clara (26.5%) and Itarare (26.3%) are the worst, against 15.7% to 16.3% in the best areas (Do Cabral, Santa Martha, Jardim Da Penha). Only neighbourhoods with at least 500 appointments are shown.

Which patient groups miss the most, and does SMS help?



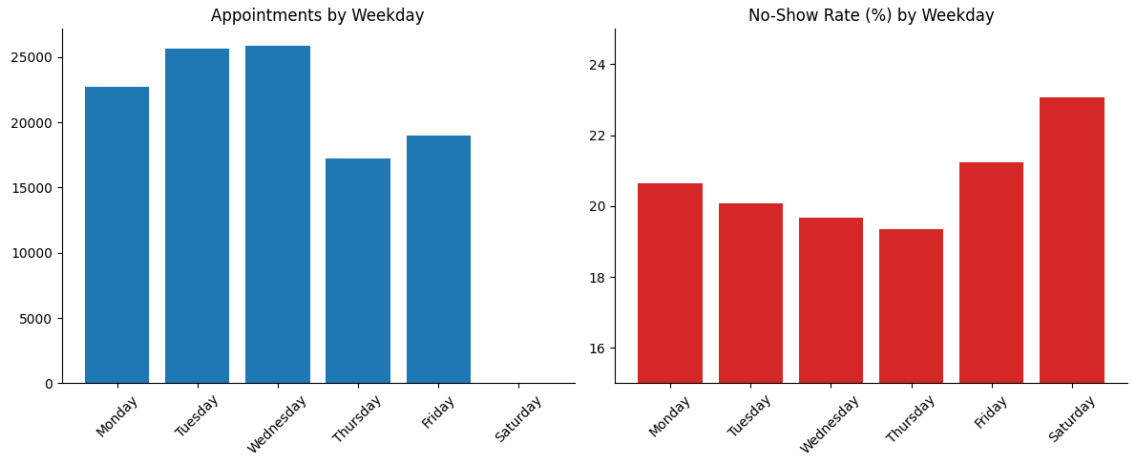
Patients aged 5 to 29 miss the most (about 24%), while patients 60 and over miss the least (15% to 16%). Overall, patients who got an SMS missed more (27.6% against 16.7%), but this is misleading, as the next chart shows.

Do SMS reminders reduce no-shows once waiting time is the same?



Yes. SMS is only sent for appointments booked ahead, which already have high no-show rates. Comparing within the same waiting band, SMS lowers no-shows by about 6 to 8 points for waits over a week (for example 36.9% to 29.8% for 15 to 30 days). No SMS were sent for same-day bookings.

Which weekday has the most appointments and no-shows?

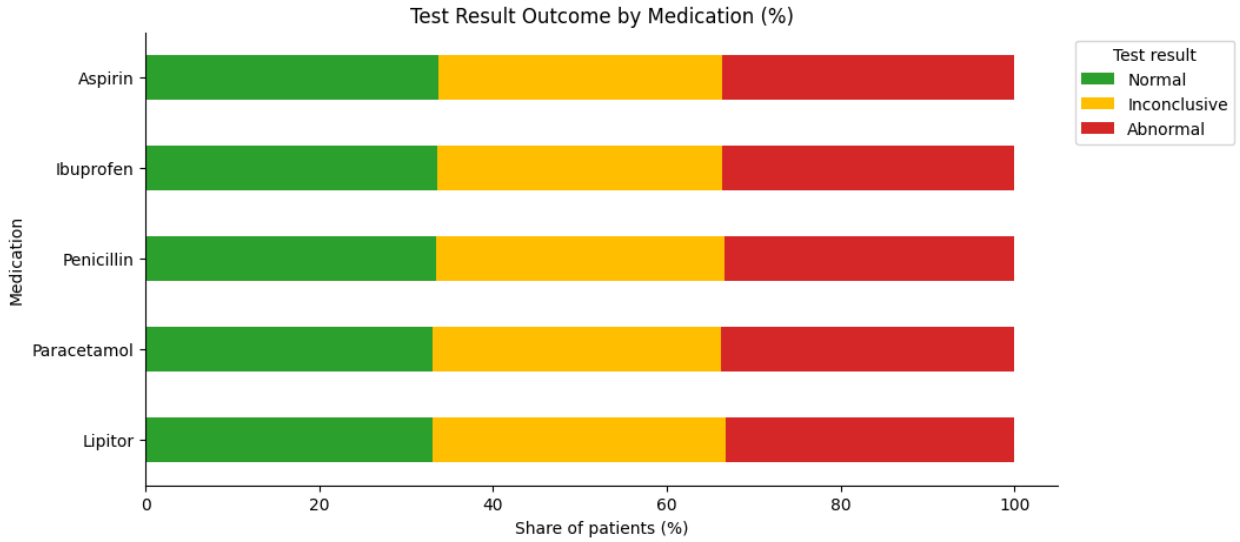


Appointments peak Tuesday and Wednesday (about 25,700 each) and are lowest on Thursday and Friday. Friday has the highest no-show rate of the main weekdays (21.2%); Saturday has only 39 appointments.

11. Treatment and Patient Outcome Analysis

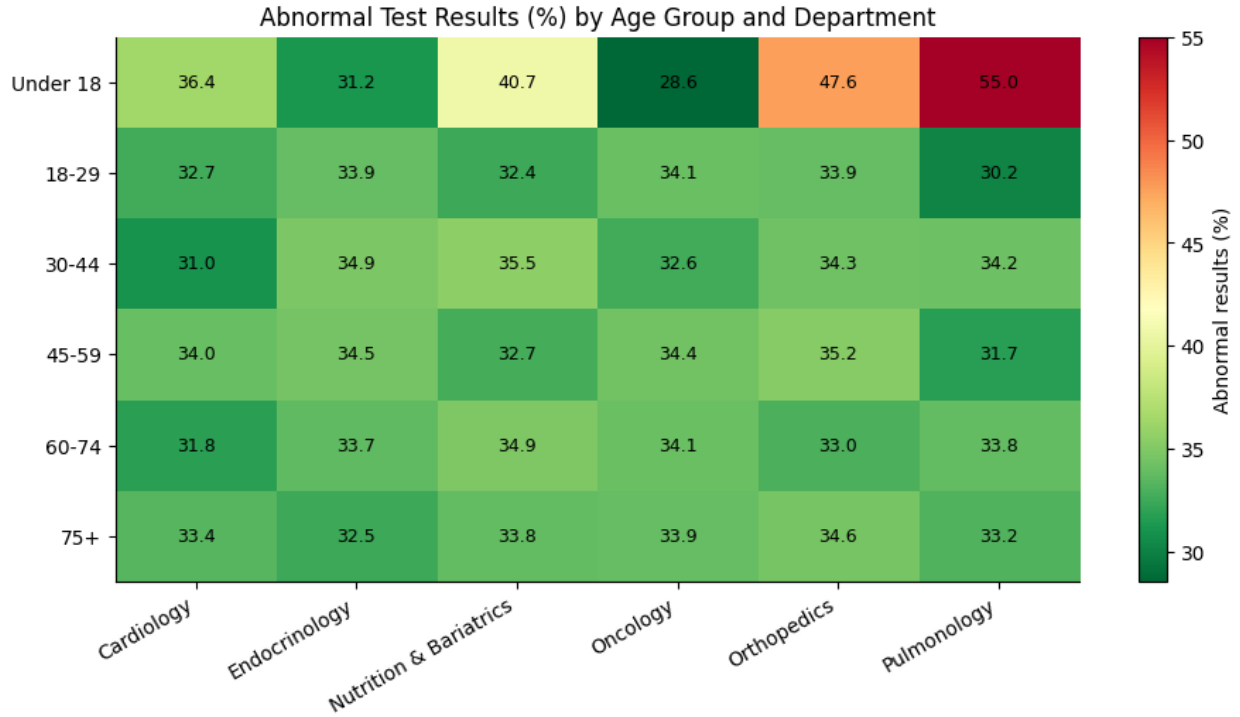
Test results (Normal, Inconclusive, Abnormal) are used as the patient outcome. Overall, 33.4% of results are Normal, 33.5% Abnormal and the rest Inconclusive.

Which treatments (medications) show the best outcomes?



All five medications have almost the same outcome mix (Normal between 33.1% and 33.7%), so no medication is clearly better in this data. Aspirin has the highest Normal rate (33.7%) and Lipitor the lowest (33.1%), a difference too small to act on.

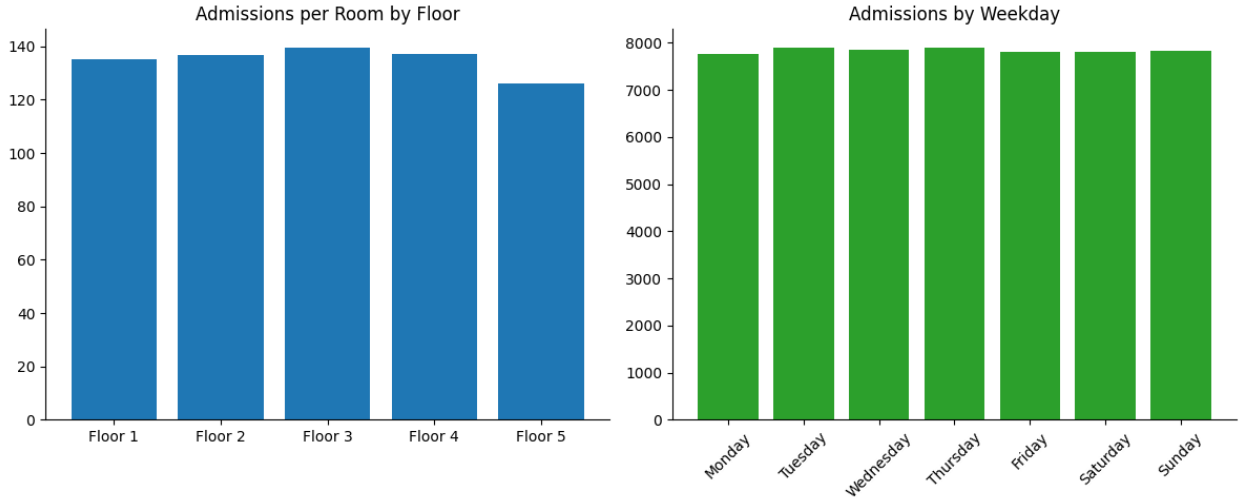
Is there a link between demographics, diagnosis and outcomes?



Abnormal results range from 28.6% to 55.0% across age group and department cells, but the highest and lowest cells are in the small Under-18 group (116 patients), so those values are unreliable. Among the adult groups the rates stay close to the overall 33.5%.

12. Hospital Resource and Operational Utilisation

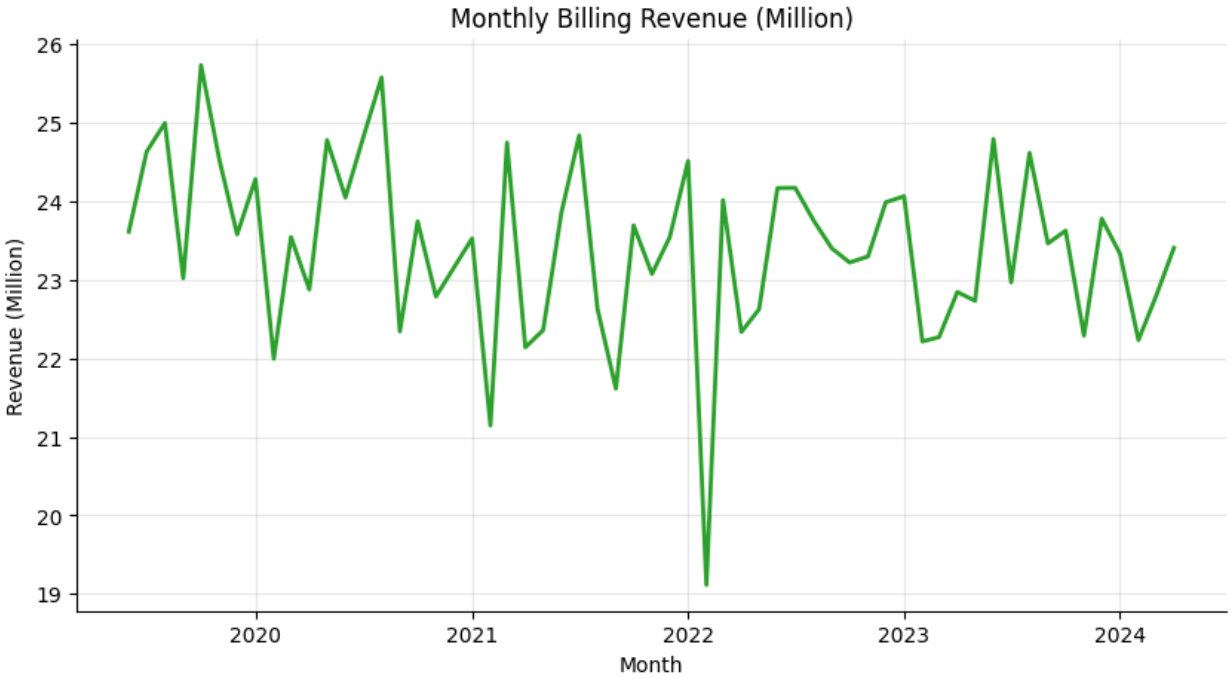
How are rooms, floors and weekdays used?



Floors 1 to 4 each handle about 135 to 140 admissions per room, so room use is balanced. Floor 5 holds a single room (500) with 126 admissions. Admissions by weekday are spread evenly. Average stay is about 15.5 days for elective, emergency and urgent admissions alike, which is long and worth reviewing as a discharge-planning opportunity.

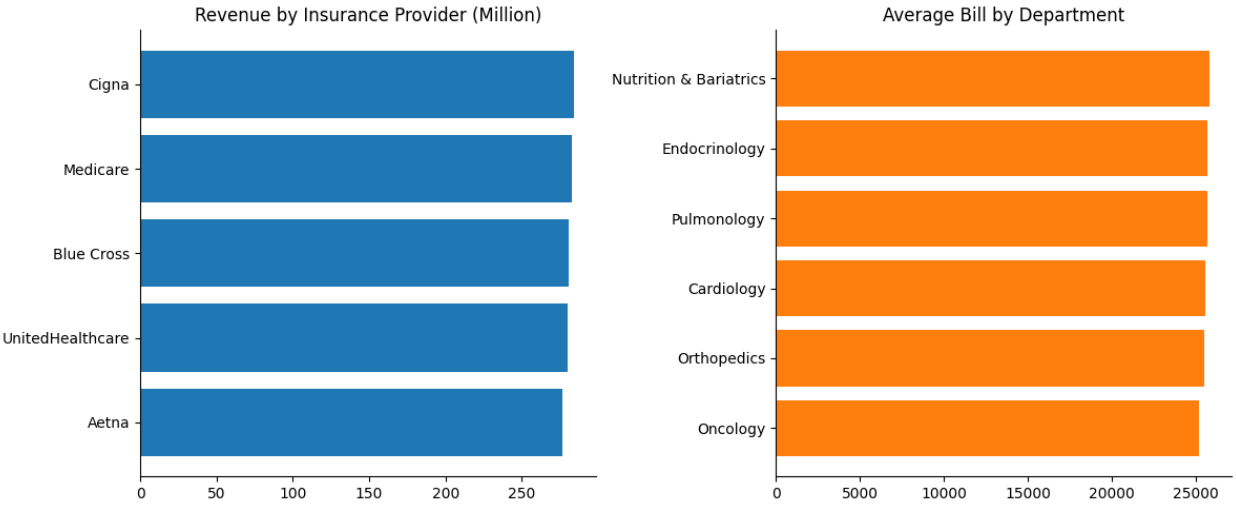
13. Billing and Revenue Analysis

What are the monthly billing and revenue trends?



Revenue is about 23.4 million per month and stable. Yearly totals are 284.0 M (2020), 277.2 M (2021), 278.6 M (2022) and 279.7 M (2023). 2019 (187.5 M) and 2024 (97.2 M) are partial years. Total revenue is 1,404.1 M.

Which patient and treatment categories cost the most?



Insurance providers contribute almost equally (Cigna 284.3 M, Medicare 282.9 M, Blue Cross 280.4 M, UnitedHealthcare 279.9 M, Aetna 276.5 M). Average bills are also similar across departments, admission types and medications (about 25,400 to 25,800). Length of stay and age have no correlation with the bill (-0.005 and -0.003), and 36.3% of revenue comes from bills over 40,000.

14. Dashboard Design (Power BI)

The dashboard (Healthcare_Dashboard.pbix) has five pages, 25 chart and table visuals, KPI cards and slicers, built on 18 DAX measures (also listed in Healthcare_Clean_Data.xlsx, sheet DAX_Measures). Every visual answers one business question.

Page	Visuals (business question)
Executive Overview	5 KPI cards; monthly admissions and revenue lines; admissions by department; revenue drill-down; revenue
Patients & Demographics	Age x gender; diagnosis by admission type; age x department abnormal-result matrix; top 10 doctors
Appointments & No-Shows	No-show by wait band, weekday, age group; weekly trend; top 10 neighbourhoods; SMS vs no SMS
Operations & Billing	Bed-days by department; floor usage; outcome by medication; average bill; department scorecard
Department Detail	Hidden drill-through page: KPIs, age/gender, outcomes, top doctors, admission types

- Slicers: year, department, gender, age group, admission type, insurance, neighbourhood, SMS reminder, weekday.
- Drill-down: year to quarter to month on the revenue chart; department to admission type on the admissions chart.
- Drill-through: right-click a department to open the Department Detail page (a Back button returns).
- Tooltips: extra measures (average bill, stay, outcome rate, appointment count) added to the hover tooltip.
- Conditional formatting: no-show bars coloured green/amber/red (20% and 25% thresholds), colour-scaled abnormal-result matrix and outcome column.

15. Executive Summary

The hospital handled 54,860 admissions worth 1,404.1 million in billing, with an average stay of 15.5 days. Its outpatient appointments show 110,514 bookings and a 20.2% no-show rate, so about one in five slots is wasted. Volume, revenue and cost are very evenly spread across departments, diagnoses, insurers and medications, so no department is overloaded. The real operational problem is appointment waiting time.

Key Findings

- Waiting time is the biggest driver of no-shows: 4.6% for same-day against 33.0% for waits over 30 days.
- 65% of appointments are booked ahead and only 34.9% are same-day, so most slots are exposed to the higher no-show risk.
- SMS reminders help: for waits over a week they cut no-shows by 6 to 8 points.
- Patients aged 5 to 29 miss the most appointments (about 24%); patients 60 and over miss the least (15% to 16%).
- Neighbourhood matters: no-show rates range from 15.7% to 28.9%.
- Admissions, stay (about 15.5 days) and bills are almost identical across departments, so there is no clear high-cost or underperforming department.
- Medication outcome rates differ by less than 1 point, so the data shows no clearly better treatment.

Actionable Recommendations

- Cut booking lead time: offer more same-day and next-day slots, and set a target of no more than 7 days for routine bookings.
- Overbook carefully on long-lead bookings: the 30%+ no-show rate for 15+ day waits means about 1 in 3 of those slots will be empty.
- Send SMS reminders for every appointment booked more than a week ahead, and add a second reminder 24 hours before.
- Target Santos Dumont, Santa Clara, Itarare and Jesus De Nazareth with phone calls or transport help, and copy practices from low no-show areas.
- Run youth and young-adult reminders through the channels they use (app, messaging), since they have the highest miss rate.
- Review the 15.5-day average stay with discharge planning; a one-day cut would free about 11,000 bed-days per year (54,860 stays over 5 years).
- Fix data capture: record department, doctor ID, cancellations and real treatment outcomes so a future dashboard can compare doctors and treatments properly.

Areas Needing Immediate Attention

- Long waits (15+ days) with a 33% no-show rate.
- High-miss neighbourhoods and the 5 to 29 age group.
- Missing data fields that block doctor-level and treatment-level decisions.